

SEIF ALDEEN SAEED AHMED

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Mechatronics and Robotics Engineering student with hands-on experience in robotics competitions, automotive systems, and suspension design. Vice Captain of Spark X Racing Team and former Class Leader. Skilled in CAD, control systems, and diagnostics, with strong leadership, communication, and problem-solving abilities.

EDUCATION

Helwan National University 2024 - 2028

B.Sc. in Mechatronics and Robotics Engineering

- CGPA: 3.65 / 4.0 (**Ranked 3rd**)
- **Class Leader (2024 – 2025).**

EXPERIENCE

Intern Aug – Sep 2025

Citroën Service Center

- Learned mechanical maintenance procedures and automotive fault detection using Diagbox software.
- Practiced error clearing, module coding, and diagnostic workflows.
- Observed maintenance tasks such as AC evaporator replacement and timing belt installation.
- Gained exposure to suspension components and other maintainable vehicle parts.

Vice Captain – Suspension System Member Feb 2025 - Present

Spark X Racing Team

- Created CAD models using Autodesk Inventor.
- Calculated suspension parameters with OptimumKinematics.
- Built a quarter car model in Simulink to analyze dynamic behavior.
- Coordinated between technical and non-technical subteams, and contributed to team representation and recruitment

Vice Head of Media Apr 2025 – Present

Hult Prize Community – Helwan National University

- Leading the Media team to manage social media presence, branding, and event coverage.
- Supporting recruitment campaigns and promotional activities.

Social Media Manager Jul 2022 - Jul 2023

H Technology – Computer Hardware Store

- Managed the brand's social media presence, boosting engagement and visibility.
- Designed and scheduled promotional content tailored to the tech audience.
- Responded to customer inquiries and supported online communications.
- Improved branding consistency through creative visual content.

VOLUNTEERING

Organizer&Speaker – Orientation Day 2025 Sep 2025

Robotics & Mechatronics Engineering Dept. – Helwan National University

- Spoke to 350+ students about academic tips, learning resources, and competitions.
- Showcased personal projects (**Gesture Control Robot, Capture the Flag Robot**).
- Introduced **SparkX Racing Team** and emphasized the value of technical skills.
- Collaborated with organizers to ensure a successful event with faculty attendance.

Robotics & Mechatronics Engineering Dept. – Helwan National University

- Delivered a workshop for Mechatronics & Robotics students on building professional LinkedIn profiles, networking, and showcasing projects.
- Engaged 70+ students and encouraged them to take immediate steps to improve their online presence.

COMPETITIONS

Auxilio Competition – 1st Place Winner**IEEE HSB | Capture the Flag Robotics Challenge**

- Led the control system and collaborated with the mechanical team to design and implement a fully functional robot for a complex Capture the Flag challenge.
- The project included developing autonomous navigation logic, obstacle avoidance, and precise control strategies to adapt to a dynamic maze environment.
- Faced and solved multiple real-world challenges such as motor synchronization issues, ultrasonic sensor calibration, and handling unexpected maze conditions.
- Strengthened teamwork, problem-solving, and rapid prototyping skills while delivering a high-performing robot that **achieved 1st place in the competition**.

RoboSoccer Competition – Participant

- Took part in a robotic soccer challenge, where the goal was to design and program robots capable of detecting, chasing, and scoring goals with an autonomous ball-tracking system.
- Contributed to the control system and strategy, ensuring smooth coordination between movement, ball detection, and decision-making.
- Faced key challenges such as sensor noise, motor response delay, and maintaining stability during fast-paced motion, which required iterative debugging and testing.
- Strengthened my understanding of real-time robotics, teamwork under competition stress, and adaptive problem-solving in dynamic environments.

PROJECTS

- **Suspension System Design (Formula Student)** – Full CAD, OptimumKinematics, and Simulink simulation.
- **Maze-Solving Robot (Auxilio 2025)** – Autonomous navigation, maze-solving logic, and sensor integration
- **Gesture Control Robot** – Designed and implemented a robot controlled through hand gestures using sensors and microcontrollers.
- **Quick Return Mechanism (QRM) Shaping Machine Model** – Designed and built a mechanical model demonstrating the Quick Return Mechanism used in shaping machines, for educational demonstration.

TECHNICAL SKILLS

- **Robotics & Control:** Arduino, ESP32, Embedded Systems, Robotics Control
- **Design & Simulation:** CAD (Autodesk Inventor), OptimumKinematics, Simulink Modeling
- **Automotive:** Diagnostics (Diagbox), Mechanical Maintenance, Suspension Design
- **Other:** Electronics Prototyping, Social Media Management, Graphic Design

SOFT SKILLS

- **Leadership & Team Management**
- **Problem-Solving & Analytical Thinking**
- **Working Under Pressure (Competition Experience)**
- **Communication & Collaboration**
- **Public Speaking & Presentation**

LANGUAGES

- **Arabic:** Native
- **English:** Fluent (C1)